

Type	Disease/ Pest/ Condition (causative organism)	Affected Bee Age/ Casts	BROOD Comb				BROOD (Larvae/Pupae) Appearance						ADULT BEE Appearance/ Behavior										
			Spotty pattern	Perforated caps	Other	"Bold" Brood	Chewed Brood	COLOR (discoloration from white)	Position in cell	Texture	Hard Scale (dried dead brood)	Smell Of Brood/ Hive	Shortened abdomen	Deformed wings	"K" wings, disjointed	Hairless/shiny, black, greasy	Twitching	Trembling/ Shaking	Paralysis	Crawling/ Unable to fly	Decreased adult population	Dysentery	Other
Bacterial Disease	American foulbrood (AFB) <i>Paenibacillus larvae</i>	Older sealed <u>brood</u> , usually, prepupae or young pupae. Adults are carriers of disease.	✓	✓	Sunken, dark cappings	✓	✓	Diseased larvae and then prepupae yellow to brown, darkening with time.	Melted larvae/pre pupae lie flat on the bottom of the cell	Goopy, roping up to 1.5" (2 - 3 cm)	Brown, black scale, <u>hard to scrape out</u> . Pupal tongue may be visible sticking up	Strong foul, rotten "fish-like" odor									✓		
	European foulbrood (EFB) <i>Melissococcus plutonius</i>	Young unsealed <u>larvae</u> at the "C" stage of development. Larvae often die before being capped.	✓	✓		✓	✓	Larvae discolored: off-white/yellow to brown with chalky white midgut and silvery tracheae visible through body wall	Twisted larvae	Melted, watery & grainy, brown pupae can rope up to 0.5", 1 cm	Easily scraped out, rubbery	Varies from normal to sour, vinegar smell									✓		
	Spiroplasma <i>Spiroplasma apis</i> , <i>S.melliferum</i>	<u>Adult</u> honey bee and <u>brood</u>	✓					Discolored larvae	Twisted larvae, in irregular position			Normal					✓	✓	✓	✓			Neurological problems-disoriented, weak.
Fungal Disease	Chalkbrood <i>Ascosphaera apis</i>	Usually older <u>larvae</u> , gets "mummified", visible in comb and often pulled out in front of hive	✓	✓	Chalk mummies are visible			Chalky white or gray, black, possible blue spots	Larva 'stand up' in cell	Hard smooth 'mummies', easily removed		Normal to slight musty odor, not repelling											
	Stonebrood <i>Aspergillus</i> sp.	From <u>larvae</u> and <u>pupae</u> to occasional adults, which turned in stone-hard "mummies"		✓	Sunken, dark cappings			Initial whiteish-yellow collar-like ring near head of larvae. Mummies with yellow or gray- greenish color		Body hardens post-death ('stone')	Mummy adhere to cell walls	Normal											

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Bacterial Disease	American foulbrood (AFB) <i>Paenibacillus larvae</i>	Spring through fall when brood present	Sunken, punctured cappings with dead melted larvae/pupae under. Once stirred with a toothpick, decomposed pupae strings out significantly. Confirm ropiness test with laboratory testing, as this is <u>highly contagious diseases</u> .	Through spores in infected food. Larvae less than 3 days old get infected during feeding, even with 1 spore.	Most infectious bacterial disease, with spores viable for 40 years. Sick colonies must be destroyed and by law- reported to CT state Apiary Inspector. Practice IPM. Monitor for disease - test for AFB at least in spring & fall (check brood appearance), different tests are available, including DIY. Practice healthy colony management (good nutrition, reduced stress, avoid overcrowding, prevent drifting) and good hive and equipment hygiene. Keep strong, healthy colonies. Don't buy/use old equipment or frames from unknown sources.
	European foulbrood (EFB) <i>Melissococcus plutonius</i>	Spring and early summer	Larvae dies before capping. Brood has visible tracheae and having a darkened appearance, and with twisted/distorted position in the cell.	Infection via oral-oral route when feeding young (2 days old) larvae with food contaminated with bacteria, contaminated frames and equipment	Practice IPM. Monitor for disease during brood rearing seasons- check brood appearance and test if any suspicion. 2 tests are available, including DIY. Moderate and high EFB infections can be treated with antibiotics, mild cases may clear up on its own with proper nutrition and when hygienic stock of bees are used. Use the same prevention methods as with AFB, requeen with hygienic stock.
	Spiroplasma <i>Spiroplasma apis</i> , <i>S.melliferum</i>	Late spring, summer. (warm, humid)	Shaking, disorientated adult honey bees. Heavy population loss. Bees may die away from colonies, so a low/declining population may be evident. Brood loss, colony decline. Diagnose thru laboratory testing	Unknown, possibly contaminated food, direct contact, wounds caused by mites. Weak, stressed colonies at higher risk.	Practice IPM. Maintain strong, healthy well nourished colonies. Reduce stress to colonies like overcrowding, poor nutrition or diseases. Provide good ventilation. Maintain good hive and equipment hygiene. Rarely occurring disease, but was reported in MD.
Fungal Disease	Chalkbrood <i>Ascosphaera apis</i>	Spring, early summer (wet, cold)	Mummies from white to dark in color at the hive entrance and bottom board, scattered throughout brood cells.	Oral-oral route when feeding larvae. Spreads through weak or stressed colonies, especially in cold damp conditions	Medium risk to colony, with proper treatment colony may recover. Practice IPM. Keep strong, healthy colonies, avoid overcrowding, maintain good hive and equipment hygiene (disinfect hive components with 1:10 bleach solution). Quarantine sick colony until full recovery. Discard infected brood comb, spores can survive for years. Ensure good nutrition and ventilation , reduce hive moisture. Requeen. Add Hive Alive supplement to the supplemental feed.
	Stonebrood <i>Aspergillus sp.</i>	Late spring, summer (warm, humid)	Restless, crawling adult bees with swollen abdomens that harden at death. Yellowish or greenish hard mummies.	Via spores in contaminated pollen, nectar or on foragers body. In humid warm conditions	Practice IPM. Keep strong, healthy colonies, maintain good hive and equipment hygiene, provide good ventilation, reduce hive moisture. Requeen.

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	Nosema <i>Nosema Ceranae</i> , <i>Nosema apis</i>	Affects the guts of <u>Adult bees</u> , all casts										Normal, foul, sour if dysentery present								✓	✓	Only in N. apis	Weak bees, not taking suppleme nt feed s
Viral Disease	Sacbrood Virus (SBV)	<u>Brood</u> , usually larvae. Sealed prepupae, need to be uncapped to be discovered	✓	✓	Capping s can be sunken			Larvae starts white, turn greyish to yellow, then brown black with head darker when dried out	larvae lay flat on bottom of the cell with head up	Larvae 'Sac- like', tough membrane filled with fluid, easily removed from cell	Dried scale rough, brittle, easily scraped out	Normal to slightly sour odor											Asymptom atic but poor foragers
	Deformed Virus (DWV)	May affect <u>all</u> life stages without clear symptoms. Noticeable on <u>pupae</u> and <u>adults</u> .	✓	✓	Pupae may die under capping s	✓	✓					Normal	✓	✓		✓		✓		✓	✓		
	Chronic Bee Paralysis Virus (CBPV)	It may affect <u>all life stages</u> without apparent symptoms, but mainly <u>adult</u> bees										Normal	✓		✓	✓		✓	✓	✓	✓	✓	Bees can be defensive
	Acute Bee Paralysis Virus (ABPV)	It may affect <u>all life stages</u> without apparent symptoms, but mainly <u>adult</u> bees	✓	✓		✓	✓					Normal				✓	✓	✓	✓ ra pid	✓	✓ rapi d		Disorienta tion
	Israeli Acute Paralysis Virus (IAPV)	It may affect all life stages without apparent symptoms, but mainly <u>adult</u> bees	✓	✓		✓	✓					Normal				✓	✓	✓	✓	✓	✓		Disorienta tion

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	Nosema <i>Nosema Ceranae, Nosema apis</i>	Late winter, spring (n. apis, all seasons (n.ceranae))	Affects digestive system of bees. Colonies appear sluggish, display poor population growth & honey production (less effective foraging). Queen supersedure increases in highly infected colonies. Dysentery in front of hive, crawling bees (N. apis only). Shortens lifespan of bees.	Contaminated food and water. Oral route through trophallaxis, grooming, cleaning contaminated cells	Most common, widespread disease posing medium risk to colonies, which usually recover with proper treatment. Practice IPM. Monitoring should be done in spring, before supering and fall, after removing supers. Provide good nutrition. Practice good hive and equipment hygiene. Replace old combs. Provide good ventilation (reduce humidity). Requeen. Treatment is available, you can also add HiveAlive supplement to a feed to help bees resist Nosema infections.
Viral Disease	Sacbrood Virus (SBV)	Spring, early summer	Discolored larvae, watery, granular in a tough, sac-like membrane. Dead prepupae need to be uncapped to be discovered (never turn into pupae). Bees usually remove diseased dead brood.	Via oral route when nurse bees feed larvae. Also by contaminated food or hive equipment.	Common viral disease. Practice IPM. Monitor for symptoms. Keep strong, healthy, well nourished colonies substitute feeding in times of dearth. Maintain good hive and equipment hygiene including cleaning tools and hive equipment and replacing old comb every few years with new one. Reduce colony stress, with close monitoring for varroa mites and other pests and diseases.
	Deformed Virus (DWV)	With peak varroa mites infestation	Bees with shriveled wings, smaller, shrunken abdomens. Bees have stunted growth and may be smaller and weaker, leading to reduced ability to forage and perform other tasks. There is dead pupae. Virus is vectored by Varroa mites.	Transmitted <u>mainly by</u> varroa mites when feeding on bees. Also by contact with infected bees or contaminated food. Can be passed from queen to the offspring. Can kill the colony	Practice IPM. Control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Use mite resistant or hygienic bee stock. Multiple treatments are available for varroa mites. Support colony health by ensuring good nutrition, avoid stressors like overcrowding, provide good ventilation. Practice good hive and equipment hygiene. Requeen with varroa resistant stock.
	Chronic Bee Paralysis Virus (CBPV)	Late spring to summer	Trembling, crawling, adult bees, unable to fly, crawling on the ground near the hive. Some bees loose hair appearing dark, shiny & greasy. Infected bees can cluster together. High mortality weakens colony significantly. PCR testing is available.	Spreading via direct contact between bees, contaminated food or hive surfaces. Varroa mites can spread CBPV	Practice IPM. Monitor for symptoms. Control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Keep strong, healthy, well nourished colonies, maintain good hive and equipment hygiene and reduce colony stress. Avoid overcrowding have good ventilation. Requeen with mite resistant or hygienic bee stock. Multiple treatments are available for varroa mites.
	Acute Bee Paralysis Virus (ABPV)	with high varroa mites infestation	Trembling, crawling, adult bees, die quickly, accumulating on the hive floor or outside the hive. Sudden decrease in adult population causing colony collapse.	Transmitted <u>mainly by</u> varroa mites, also by contact with infected bees. Contribute to <u>colony collapse!</u>	Practice IPM. Monitor for symptoms. Control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Keep strong, healthy, well nourished colonies, maintain good hive and equipment hygiene and reduce colony stress. Avoid overcrowding have good ventilation. Requeen with mite resistant or hygienic bee stock. Multiple treatments are available for varroa mites.
	Israeli Acute Paralysis Virus (IAPV)	with peak v. mites infestation	Trembling, crawling, adult bees, die quickly, accumulating on the hive floor or outside the hive. Sudden decrease in adult population.	Transmitted <u>mainly by</u> varroa mites, also can be transmitted horizontally from queen to the offspring. Contribute to <u>colony collapse!</u>	Practice IPM. Monitor for symptoms. Control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Keep strong, healthy, well nourished colonies, maintain good hive and equipment hygiene and reduce colony stress. Avoid overcrowding have good ventilation. Requeen with mite resistant or hygienic bee stock. Multiple treatments are available for varroa mites.

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	Kashmir Bee Virus (KBV)	It may affect all life stages without apparent symptoms, but mainly <u>adult</u> bees.	✓	✓		✓	✓					Normal				✓	✓	✓	✓	✓		Often asymptomatic	
	Black Queen Cell Virus (BQCV)	<u>Queen pupae</u>			Queen cell wall dark brown, black			Larvae yellow darkens after death.		Tough 'Sac-like' filled with fluid		Normal			✓								
Parasites	Tracheal Mite <i>Acarapis woodi</i>	Adult bees										Normal			✓ possible		✓ possible		✓	✓	✓		
	Varroa Mite <i>Varroa Destructor</i> Parasitic Mite Syndrome (PMS)	<u>Adult, prepupa, and pupae/ All casts,</u> prefer Drone brood for reproduction	✓	✓	Mites feces (guanin e) visible in cells	✓	✓	Larvae / pupae turns brown with decay	Larva 'stand up" in cell, twisted	Goopy pupae, doesn't rope	Hard scale easily scraped out	Normal	✓	✓	✓	✓			✓	✓		Strong colonies may collapse late summer, early fall	
	Small Hive Beetles (SHB) <i>Aethina Tumida</i>	Brood	✓	✓	SHB eggs & larvae in comb	✓						Fermented smell of spoiled nectar/ honey											

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	Kashmir Bee Virus (KBV)	with peak v. mites infestation	Disease is often asymptomatic, unless colony is under stress, then similar to ABPV. Decreased adult population.	Transmitted <u>mainly by</u> varroa mites, also by contact with infected bees.	Practice IPM. Monitor for symptoms. Control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Keep strong, healthy, well nourished colonies, maintain good hive and equipment hygiene and reduce colony stress. Avoid overcrowding have good ventilation. Requeen with mite resistant or hygienic bee stock. Multiple treatments are available for varroa mites.
	Black Queen Cell Virus (BQCV)	Early spring & summer during queen rearing	Dead queen pupae with black cell walls. Associated with Nosema spp. infections.	Via oral route when nurse bees feed larvae. Associated with Nosema spp. infections.	Practice IPM and good hive and equipment hygiene and healthy colony practices. Prevent robbing, drifting, open feeding. Monitor for Nosema. Risk to queen rearing operations.
Parasites	Tracheal Mite <i>Acarapis woodi</i>	Winter, spring	Crawling bees at the entrance, unable to fly, sometimes in large numbers. Infected colony dwindle and dies overwinter. Microscopic examination of trachea.	By contact with infected bees. Infected are usually young bees, less than 4 days old	Rare due to resistant bee strains and treatments. Practice IPM, Monitor for disease. Keep strong, healthy colonies. Menthol treatments are available.
	Varroa Mite <i>Varroa Destructor</i> Parasitic Mite Syndrome (PMS)	With peak varroa mites infestation, late summer, fall	On occasion you may spot reddish female mites adhering to an adult bee on or near the abdomen on the underside. Seeing mites on bees may indicate already high infestation. You can find whitish spots (guanine- mite feces) inside recently emerged honey bee cells. With heavy mite infestation you may see increased symptoms of viruses and other diseases present. Monthly testing for mites is the only way to assess the mite infestation rate in the colony.	During foraging, robbing, drifting, open feeding. By contact with infected bees, with brood frames exchanges	Practice IPM. Monitor (monthly testing) & control Varroa mite population, prevent robbing and drifting. Don't open feed during varroa outbreaks. Use mite resistant or hygienic bee stock. Follow healthy colony practices (good nutrition, reduced stress, proper spacing) and hive and equipment hygiene. Various varroa mites treatments are available. Parasitic mites Syndrome (PMS) its a set of symptoms associated with high mite infestation along with viruses they vector. It can cause colony collapse in 1-4 weeks, usually in late summer/early fall (August), and may affect the most robust colonies. Learn the biology of varroa mites to be able to help your colonies.
	Small Hive Beetles (SHB) <i>Aethina Tumida</i>	Mainly in summer & fall	Presence of adult SHB or larvae in the hive and screened bottom board. Fermented honey may be leaking in high infestations.	Adult SHB enter weak/unprotected hives. Larvae will finish development cycle in the ground around/under the hive if allowed.	Practice IPM. Monitor for symptoms. Keep strong, healthy, well nourished colonies, maintain good hive & equipment hygiene and reduce colony stress. Avoid overcrowding and maintain good ratio of bees/hive space. Manage space under and around hives to prevent SHB development (mechanical, biological or chemical barrier). Use mechanical traps inside the hives, manage entrances in smaller/weaker colonies. Remove burrcomb & keep hive & frames in good conditions. Monitor supplemental feeding to what bees can consume in short periods. Chemical treatment is available. Learn the biology of SHB to be able to help your colonies.

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Pests	Wax Moth <i>(Galleria mellonella, Achroia grisella)</i>	Brood		✓	Wax larvae in comb	✓						Normal											
	Bee Louse <i>Baula Coeca</i>	Adult bees, all casts										Normal											Bees w./ brown lice on top
	Ants	Adult bees										Normal											Bees can be defensive
	Wasps, hornets	Adult bees										Normal								✓			Bees are defensive
	Mouse, rats, shrew	Adult bees and brood										Normal											Bees can be defensive
Pests	Raccoon, skunks, badgers	Mainly adult bees										Normal								✓			Bees can be defensive
	Bears	Adult bees and brood										Normal											

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Pests	Wax Moth (<i>Galleria mellonella</i> , <i>Achroia grisella</i>)	Summer, Fall	Silk tunnels, webbing or cocoons in comb & wooden hive parts. Chewed comb, uncapped brood. Larvae in the comb. Dark droppings of wax moths left behind larvae - in the comb or on screened bottom board.	Reusing frames and equipment from infected hives,	Practice IPM. Maintain strong colonies , keep good ratio of bees/ hive space. Keep hive equipment in good condition- minimize gaps and cracks. Keep good apiary hygiene. Properly store frames and hive equipment - freeze unused frames for 24-48 hours to kill eggs and larvae before storing. set Wax moth traps. Learn the biology of Wax moths to be able to help your colonies.
	Bee Louse <i>Baula Coeca</i>	Mainly in summer & fall	Tunnels under honey cappings, bees with brown lice on the top of thorax or abdomen.	Direct contact between bees or hive components. Robbing, drifting, swarming.	Rare, minor concern to colonies. Practice IPM. Maintain strong colonies and good hive and equipment hygiene to prevent transfer of lice. Avoid sharing equipment between colonies without proper cleaning. Prevent robbing and drifting. Quarantine new colonies before introducing them to the apiary- if possible. Freeze infested comb.
	Ants	All seasons	Ant trails leading to the hive, ants entering hives through cracks, nesting material, eggs inside or near the hive.	N/A	Nuisance, but could be a thread to colonies, as they may steal honey & cause stress to the colony & possibly absconding. Elevate hives & install ant barrier. Keep hive equipment in good condition (no cracks, wholes). Keep apiary clean of any bee products, debris & excessive vegetation, avoid honey/sugar syrup spills. Use deterrents like diatomaceous earth, cinnamon or use bait traps.
	Wasps, hornets	Late summer, fall	Hives are robbed of honey, bees are attacked near entrance. Lack of honey, wax debris on screened bottom board, shredded edges of comb. Dead bees, partially eaten around hives. Wasps, hornets hovering near hive entrance.	N/A	Practice IPM. May pose significant threat to colonies, especially during dearth period. Reduce entrances /use robbing screens during dearth periods. Keep healthy & strong colonies. Keep apiary clean of any bee products, debris and excessive vegetation, avoid honey/sugar syrup spills. If wasps/hornets are known problem in your apiary - set traps around apiary in early spring to eliminate queen and prevent colonies from building. Remove nearby nests if needed.
	Mouse, rats, shrew	Cold season	Nest material inside the hive, chewed entrances, frames, comb and hive parts. Droppings in the hive, disturbed /dead bees with no abdomen (shrews).	N/A	Practice IPM. Nuisance for beekeepers. Install mouse guards in early fall-spring, or make sure the entrance hole is no taller than 3/8", use metal entrance reducers. Elevate hives above ground. Store hive equipment in safe place, frames & comb should be stored in sealed, rodent-proof containers. Keep apiary clean of any bee products, debris & excessive vegetation, avoid honey/sugar syrup spills. Trap/ deter rodents.
Pests	Raccoon, skunks, badgers	All seasons	Scratches or bite marks on hive entrance, landing board or screen bottom boards. Hives tipped over or top cover removed. Scattered debris. Digging signs around hives.	N/A	Nuisance for beekeepers. Install electric fence with wires near the ground with reliable charger, strap the hives. Elevate hives at least 18" above the ground, install metal entrance reducers. Keep apiary clean of any bee products, debris and excessive vegetation, avoid honey/sugar syrup spills. You can install motion activated lights around apiary.
	Bears	Spring and Fall	Damaged, knocked over hives, claw or bite marks. Bear can kill entire colony or cause it to abscond.	N/A	High risk to colonies. Practice IPM. Install electric fence with reliable chargers, strap the hives. Keep hives in safe distance from the fence- at least 3-4 feet. Keep apiary clean of any bee products and debris. Store frames and unused hive equipment in safe location.

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Other	Paralysis (Viruses: CBPV, ABPV, IAPV, KBV), Toxic plants, Poisoning	Adult bees										Normal				✓	✓	✓	✓	✓	✓		Disorient ed or dead bees around hives
	Poisoning (pesticides exposure)	Adult bees										Normal					✓	✓	✓	✓	✓		Disorienta tion
	Poor nutrition (pollen, nectar)	Adult bees and brood	✓		Low amount s of Royal jelly		✓					Normal	✓					✓			✓	✓	Weak bees
	Drone laying Queen	Brood	✓		Only drone brood produc ed							Normal											Only drones are produced
	Laying Workers	Brood	✓	✓	Multipl e eggs/lar vae per cell		✓		on the cell walls rather than center			Normal											Only drones are produced
	Chilled Brood	Brood	✓	✓	Dead larvae in comb		✓	Grayish larvae, looks like Sacbrood, pupae dead under capping				Unpleasa nt, decaying brood											

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Other	Paralysis (Viruses: CBPV, ABPV, IAPV, KBV), Toxic plants, Poisoning	Foraging seasons	Bees with neurological problems like trembling, jerky movements, disoriented, crawling, unable to fly, near entrance or on the ground. Bees with shiny, greasy, hairless bodies. Weak, lethargic or dead bees. Decreased population and foraging activity.	A lot of viral infections are associated with varroa mite infestations. Spreading through direct contact between bees, contaminated hive components, comb and beekeeping tools, trophallaxis and grooming.	High risk to colonies. Practice IPM. Strengthen colony health and practice good hive management. Control varroa mites infestations, clean hive tools and feeders frequently, avoid sharing frames from infected colonies, keep hive components in good condition, rotate old comb. Reduce colony stress: avoid overcrowding, provide good nutrition, prevent robbing and drifting, keep good ventilation in the hive. Don't open feed during varroa outbreaks. Use mite resistant or hygienic bee stock. Monitor for toxic plants and pesticide use.
	Poisoning (pesticides exposure)	Foraging seasons	It depends on the pesticide exposure: chronic or acute. Bees with neurological problems, weakened or dead bees, often with extended tongue.	Contaminated food brought to hive or direct contact during foraging	Monitor for toxic plants and pesticide use. Avoid use of pesticides around your apiary. Provide source of clean water. Know the foraging sources in 1, 3, 5 miles radius around your apiary, learn the risk factors, "know your neighbors" and agricultural practices in foraging radius around your apiary (up to 5 miles).
	Poor nutrition (pollen, nectar)	During pollen/nectar Dearth	Reduced storage of pollen, nectar in the hive. Reduced amounts of royal jelly in cells with small larvae. Breaks in bee brood stages (small larvae are cannibalized). Weak bees.	N/A	Monitor food sources through the season, supplement feed if needed with high quality food and clean water. Plant for forage diversity through the seasons, especially during the dearth.
	Drone laying Queen	All seasons	No worker brood is produced, only drones, even in worker cells you can see drone bullet shape cappings. Drones may be smaller. Colony will dwindle & die if not addressed. New queen is poorly or not mated or old queen run out of sperm.	N/A	Perform regular hive inspections on at least 3 weeks basis. Check queen status and laying pattern at every inspection, especially with newly mated queens. You don't have to see the queen but should be able to spot the proper brood. Requeen the colony if there are issues with the brood, and no workers are produced. Strengthen the colony by adding brood frames and bees from other colonies.
	Laying Workers	Foraging seasons	No queen in the colony. Multiple eggs laid in the cells and they are not on the bottom of the cell in the center. Only drone brood is produces, even in worker cells. Colony is doomed for death if proper queen is not introduced.	N/A	Perform regular inspections on at least 3 weeks basis. Confirm colony queen-right status during every inspection, so you can intervene quickly if the queen is lost. You don't have to see the queen but should be able to spot the proper brood. Always confirm new queen status when introduced to colony, making sure queen is laying correctly. Monitor brood for proper eggs/larvae presence. Act quickly to try to save the laying workers colony, few methods are available depending on colony status.
	Chilled Brood	Early spring	Dead larvae in the cells or pulled in front of the hive. Small number of nurse bees, not covering the brood. Spotty brood pattern.	Remove/clean frames with chilled brood to prevent decay and hive contamination	Don't work with brood when weather is cold (less than 60F), keep the brood frames out of the hive for a very short period (broodnest is kept in approx. 94F). Keep the order of brood frames intact. Combine small/weak colonies or add nurse bees if needed to cover the entire brood. Reduce colony stress, limit excessive inspections, make quick inspections during warmest part of the day. Avoid windy humid conditions.

Type	Disease/ Pest/ Condition <i>(causative organism)</i>	Affected Bee Age/ Casts	BROOD Comb		BROOD (Larvae/Pupae) Appearance							ADULT BEE Appearance/ Behavior											
			Other	Perforated caps	Spotty pattern	"Bold" Brood	Chewed Brood	COLOR (discoloration from white)	Position in cell	Texture	Hard Scale (dried dead brood)	Smell Of Brood/ Hive	Shortened abdomen	Deformed wings	"K" wings, disjointed	Hairless/shiny, black, greasy	Twitching	Trembling/ Shaking	Paralysis	Crawling/ Unable to fly	Decreased adult population	Dysentery	Other
Other	Starvation	Adult bees and brood	✓		Reduce d amount of royal jelly		✓					damp, musty, decaying if bees are dead											Dead bees with heads in the cells
	Robbing	Adult bees										Normal											Dead bees around entrance
	Dysentery	Adult bees										Sour fecal smell											Bees defecating inside hive
	Overcrowding	Adult bees and brood			Lack of space for brood																		Swarm prep, fanning, stressed bees, defensive

Type	Disease/ Pest/ Condition <i>(causative organism)</i>	Season	Field Identification (symptoms and diagnosis)	Infecting/ Spreading	Prevention/ Treatments
Other	Starvation	Winter, spring, nectar dearth	Large number of dead bees on the bottom or clustered in the comb with heads in the cells. No honey/nectar sources in the hive or far away from the cluster. Brood break, weak, lethargic bees. Bees may abscond the hive.	N/A	Monitor food stores at every inspection. Bees should always have at least 2 frames with honey/nectar and at least a frame of pollen. To prepare for overwintering colony should have approximately 6-8 frames of honey (60-80lb) and 1-2 frames of pollen. Provide supplemental or emergency feed if needed. Leave sufficient honey after harvest. Understand local foraging sources and plan for the dearth periods. Combine weak, small colonies before the winter. Keep proper ratio of bees/space, condense bees to smaller box if possible.
	Robbing	Nectar dearth	Large number of bees aggressively trying to get access to a hive, checking joints and crack. Bees fighting at the entrance. Bees flying out fast with extended abdomens. Wax debris on the screened bottom board.	Spilling nectar, honey, syrup around hives, keeping hives opened for a long time during dearth season.	Avoid spilling honey/nectar/syrup around hives. Maintain good food sources in the colonies, especially during the nectar dearth. Keep inspections short, don't expose honey frames for a long time. Inspect during high foraging time, usually mid day. Keep strong colonies. Install robbing screens during high robbing risk season.
	Dysentery	Confinement, cold	Orange-brown feces inside the hive, on the frames or outside the hive. Bees are not defecating away from the hive due to cold weather or excessive fluid buildup in bee's guts	Contaminated or fermented food (honey/pollen).	Make sure bees go into winter with fully matured honey with appropriate water content. Avoid dark honeys stored for the winter as they are high in ash content and may cause diarrhea. Avoid storing contaminated or fermented honey. Provide good hive ventilation.
	Overcrowding	Nectar flow	Excessive bees in the hive, no space to lay eggs, swarm preparation (queen cells), poor ventilation- high humidity or heat inside the hive, bees fanning or bearding, behavioral stress- increased aggression or disorientation in bees.	N/A	Provide adequate space (add supers, rotate frames, harvest honey), split the Colony (create nucs or perform artificial swarm), improve ventilation, monitor brood nest for space to lay eggs, requeen if needed, Avoid overfeeding (if supplementally feeding colony).